

FYS5419/9419
lecture April 30

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$$\mathcal{L} = 1\lambda - \frac{1}{2} \lambda^T P \lambda$$

$$P = \begin{bmatrix} y_1 y_1 k(\vec{x}_1, \vec{x}_1) & y_1 y_2 k(\vec{x}_1, \vec{x}_2) & \dots & y_1 y_n k(\vec{x}_1, \vec{x}_n) \\ y_2 y_1 k(\vec{x}_2, \vec{x}_1) & - & - & - \\ \vdots & & & \\ y_n y_1 k(\vec{x}_n, \vec{x}_1) & - & - & y_n y_n k(\vec{x}_n, \vec{x}_n) \end{bmatrix}$$

$$\mathcal{L} = \sum_{i=1}^n \lambda_i - \frac{1}{2} \sum_{i,j} \lambda_i \lambda_j y_i y_j k(\vec{x}_i, \vec{x}_j)$$

$$k(\vec{x}_i, \vec{x}_j) = \vec{x}_i^T \vec{x}_j$$

$$y_i \lambda_i' = h_i'$$

$$\sum_{i,j} h_i' h_j' K(\vec{x}_i, \vec{x}_j) \geq 0$$

for each set of n
 $(h_i' \in \mathbb{R})$

then there is a mapping
 $\phi(\vec{x})$ such that $-K-$ can
 be written as

$$K(\vec{x}_i, \vec{x}_j) = \phi(\vec{x}_i)^T \phi(\vec{x}_j)$$

$$\sum_{i,j}^n h_i h_j' K(\vec{x}_i, \vec{x}_j) =$$

$$\left(\sum_{i=1}^n h_i \phi^T(\vec{x}_i) \right) \left(\sum_{j=1}^n h_j' \phi(\vec{x}_j) \right)$$

$$\geq 0$$

Can define a function

$$f(\vec{x}_j) = \sum_{i=1}^n \frac{\lambda_i y_i'}{h_i'} K(\vec{x}_i, \vec{x}_j) \quad (*)$$

In QM a state is a superposition of special functions

$$\hat{H} \phi_n(x) = E_n \phi_n(x)$$

$$\phi(x) = \sum_{n=0}^{\infty} c_n \phi_n(x)$$

$$c_n = \langle \phi_n(x) | \phi(x) \rangle$$

$$|c_n|^2 = |\langle \phi_n(x) | \phi(x) \rangle|^2$$

$$\langle \phi | H | \phi \rangle = \sum_{nm} (\langle \phi_n | c_n^*) H (| \phi_m(x) \rangle c_m)$$

$$= \sum_n |c_n|^2 E_n$$

(***)

if we compare $(*)$ and $(**)$,
 we could think of using the
 quantum system as a kernel
 machine by Tayloring
 the energy levels E_n as
 the labels $-y_i$ and
 exploit $|\langle \phi_n | \phi \rangle|^2$ as
 some type of kernel function.

$$|\phi_n\rangle \rightarrow |n\rangle$$

$$\hat{O} = \sum \lambda_n E_n |n\rangle \langle n|$$

$$[\hat{H}, \hat{O}] = 0$$

$$\langle \phi | \hat{O} | \phi \rangle = \sum_n \underbrace{\lambda_n E_n}_{O_n} |\langle n | \phi \rangle|^2$$

Mimics $f(x)$ in ~~(x)~~
if we identify

$$K_n(x) = |\langle n | \phi \rangle|^2$$

$$\langle \phi(x) | \hat{O} | \phi(x) \rangle = f(x)$$

$$x_i \longrightarrow |\phi(x_i)\rangle$$

we have to make a mapping
from x_i to a quantum
mechanical state

and assign the label y_i when we observe such a state

$$\hat{O} = \sum_{i=1}^n x_i y_i | \phi(x_i) \rangle \langle \phi(x_i) |$$

$$\begin{aligned} f(x) &= \langle \phi(x) | \hat{O} | \phi(x) \rangle \\ &= \sum_{i=1}^n x_i y_i \underbrace{|\langle \phi(x_i) | \phi(x) \rangle|}_{k(x_i, x)}^2 \end{aligned}$$

$$f(x_j) = \sum_{i=1}^n x_i y_i \underbrace{k(x_i, x_j)}_{k_{ij}}$$

$$f(x_j) = \sum_{i=1}^n \underbrace{O_i}_{\lambda_i y_i} \underbrace{K_{ij}}_{|\langle \phi(x_i) | \phi(x_j) \rangle|}$$

$$x \mapsto \phi(x)$$

Density matrix

$$\rho(x) = |\phi(x)\rangle \langle \phi(x)|$$

can show that

$$\begin{aligned} K(x_i, x_j) &= \text{Tr} \{ \rho(x_i) \rho(x_j) \} \\ &= |\langle \phi(x_i) | \phi(x_j) \rangle|^2 \end{aligned}$$

$$R_x(x) = e^{-i \frac{x}{2} \sigma_x} =$$

$$\cos\left(\frac{x}{2}\right) \mathbb{1} - i \sin\left(\frac{x}{2}\right) \sigma_x$$

$$\sigma_x = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

$$R_x(x) |0\rangle = \cos\left(\frac{x}{2}\right) |0\rangle - i \sin\left(\frac{x}{2}\right) |1\rangle$$

$$U(x) |0\rangle = |\phi(x)\rangle$$

$$f(x) = |\phi(x)\rangle \langle \phi(x)| =$$

$$\cos^2\left(\frac{x}{2}\right) |0\rangle\langle 0| + \lambda \cos\left(\frac{x}{2}\right) \sin\left(\frac{x}{2}\right) \times |0\rangle\langle 1|$$

$$\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \quad \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$$

$$- \lambda \cos\left(\frac{x}{2}\right) \sin\left(\frac{x}{2}\right) |1\rangle\langle 0|$$

$$+ \sin^2\left(\frac{x}{2}\right) |1\rangle\langle 1|$$

$$\langle f(x), f(x') \rangle = \text{Tr} \{ f(x) f(x') \}$$

$$= K(x, x') =$$

$$\left| \cos\left(\frac{x}{2}\right) \cos\left(\frac{x'}{2}\right) + \sin\left(\frac{x}{2}\right) \sin\left(\frac{x'}{2}\right) \right|^2$$

$$= \cos \left(\frac{x-x'}{2} \right)^2$$

$$f(x_j) = \sum_{i=1}^n E_i^T \text{Tr} \{ f(x_i) f(x_j) \}$$

computational model

- Map all the points in the dataset into quantum states
- compute the scalar products using a given quantum device
- invert the Gram matrix